Prajna Ranganath

Electronics and Communication Engineer | Robotics | Embedded Systems | Cyber Physical Systems
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Professional Summary

Electronics and Communication Engineering graduate specializing in Cyber Physical Systems with research experience in agricultural robotics at Cornell University. Interested in robotics, embedded systems, autonomous navigation, computer vision, and intelligent control.

Education

SASTRA Deemed University
Bachelor of Technology in Electronics and Communication Engineering
Specialized in Cyber Physical Systems
July 2022 - July 2026
CGPA: 7.9/10.0

Experience

Research Intern
Cornell University – College of Agriculture and Life Sciences (CALS)
February 2026 – June 2026

- Contributed to research in agricultural robotics through software, embedded systems, and hardware development.
- Developed robotic software and automation tools for agricultural applications.
- Designed embedded electronics and supported hardware integration and testing.
- Investigated system reliability and implemented hardware improvements.
- Collaborated on the development and evaluation of robotic research platforms.

Projects

Autonomous Robot Navigation and Path Planning

- Developed a modular ROS2 framework for autonomous navigation using occupancy grid mapping and graph-based path planning.
- Implemented graph-search algorithms, including A*, for collision-free trajectory generation with heuristic optimization in complex environments.

Dynamic PID Controller for DC Motor Speed Regulation

- Developed an adaptive closed-loop speed controller on ESP32 and STM32 using real-time encoder feedback.
- Explored soft computing techniques, including Fuzzy-PID control, reducing overshoot from ~20% to <5% and settling time to <1 s.

LiDAR-Based Environmental Mapping and Perception

- Developed a modular LiDAR perception pipeline for processing 3D point-cloud data.
- Implemented filtering, ground removal, voxel downsampling, clustering, and object-level representation for autonomous navigation.

IoT-Based Red Palm Weevil Detection System

- Developed a compact CNN model for embedded, real-time Red Palm Weevil detection using thermal imagery, achieving 89% accuracy.
- Evaluated CNN, YOLO, and Vision Transformer (ViT) architectures, optimizing the final model for deployment on a resource-constrained edge device.

Technical Skills

- **Programming Languages:** C, C++, Python, MATLAB
- **Robotics:** ROS, ROS2, Gazebo, RViz, Autonomous Navigation, Occupancy Grid Mapping, Path Planning, LiDAR Perception
- **Embedded Systems:** STM32, ESP32, Arduino, Teensy, Embedded C/C++, Sensor Integration
- **Control Systems:** PID Control, Adaptive Control, State Estimation, Motion Control
- **Computer Vision and AI:** OpenCV, Image Processing, Convolutional Neural Networks (CNNs), Vision Transformers (ViTs)
- **Electronics and PCB Design:** KiCad, PCB Design, Circuit Design, Motor Drivers, Encoder Systems, Hardware Debugging
- **Development Tools:** Git, Linux, WSL, PlatformIO, STM32CubeIDE, VS Code

Leadership and Activities

Core Member: The Electronics Club (TECS), SASTRA Deemed University

- Organized technical workshops and hands-on electronics sessions for undergraduate students.
- Supported club activities promoting practical learning in embedded systems and robotics.

Core Member: DAKSH – SASTRA Techno-Cultural Fest

- Coordinated logistics and resource management for one of the university's largest techno-cultural festivals.
- Collaborated with multidisciplinary teams to ensure efficient planning and successful event execution.

Certifications

- NPTEL – Drone Design and Control
- NPTEL – Industrial Internet of Things
- NPTEL – 5G Standard Design
- Electronics Manufacturing – Government of India

Languages

English, Kannada, Tamil, Hindi, German (A1)